Subject: Manufacturing Processes (23A0340IT)

UNIT - V : ADDITIVE MANUFACTURING

Steps in AM, classification, advantages, materials, VAT photopolymerization, extrusion-based, powder bed fusion, directed energy deposition, post-processing, applications

1. Introduction to Additive Manufacturing (AM)

Additive manufacturing (3D printing, rapid prototyping, layered manufacturing) is a process of making a part by adding material layer by layer from a 3D CAD model, as opposed to subtractive (machining) and formative (casting, forging) processes. ASTM F2792 definition: "joining materials to make objects from 3D model data, usually layer upon layer".

- Evolution: Rapid Prototyping (1987 - first SLA machine by 3D Systems, Chuck Hull) -> Rapid Tooling -> Rapid Manufacturing -> Direct digital manufacturing of end-use parts.
- Principle: solid 3D model is sliced into thin 2D layers (0.01 - 0.5 mm); each layer is built by depositing / solidifying material and bonded to the previous layer; the part grows in the Z direction.
- Near-net shape; geometric complexity is free - internal channels, lattices, topology-optimised shapes.

	Subtractive	Formative	Additive
Principle	Material removed from block	Shape by deformation / mould	Material added layer by layer
Waste	High (chips)	Low-moderate	Very low (5-10 %)
Tooling	Cutters, fixtures	Dies, patterns (costly)	None
Complexity	Limited by tool access	Limited by die	Almost unlimited
Best volume	Low-medium	High	Low, customised

2. Steps in Additive Manufacturing

- Step 1 - CAD: solid or surface model; can come from CT/MRI or 3D scanner (medical, reverse engineering).
- Step 2 - STL: surface approximated by triangles (vertices + normals); resolution set by chordal tolerance; must be a closed (water-tight) manifold.

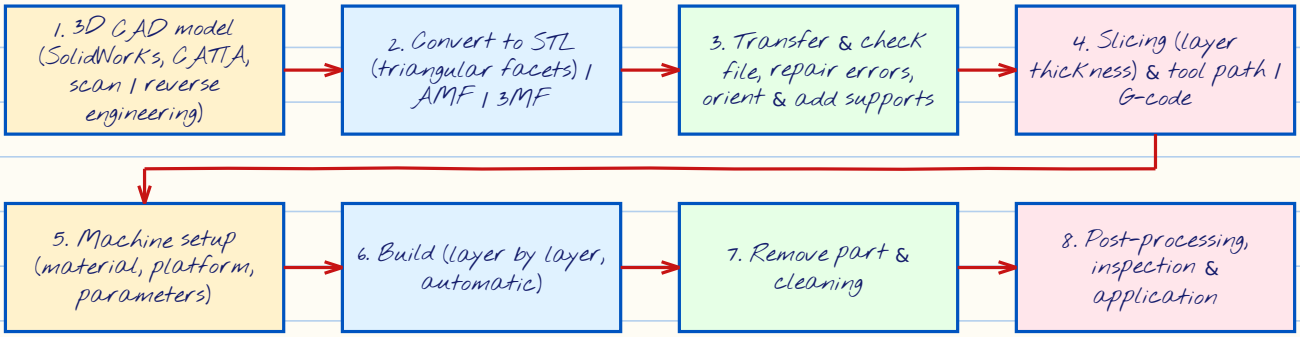


Fig 2.1 Generic AM process chain

Errors: gaps, flipped normals, overlapping triangles - repaired with Magics / Netfabb.

- Step 3 - Build orientation & supports: orientation affects build time, accuracy, strength (anisotropy: weakest in Z), surface finish and support quantity; supports hold overhangs (> 45 degrees) & resist warping.
- Step 4 - Slicing: STL cut into layers; slice thickness 20-100 microns (SLA) to 0.1-0.4 mm (FDM); produces contour & hatch paths. Staircase effect: stepped surface on curved faces - smaller layer = better finish but longer time.
- Step 5-6 - Setup & build: set temperature, laser power, scan speed, hatch spacing, layer thickness; build is automatic; build time from hours to days.
- Step 7-8: remove, clean, post-process (see section 8), inspect (CMM, CT scan), use.

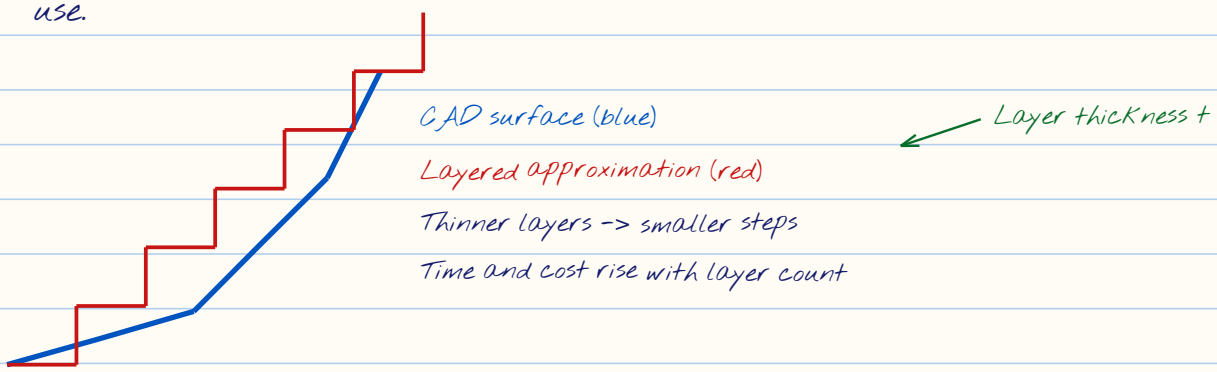


Fig 2.2 Staircase effect

3. Classification of AM Processes

ASTM F42 / ISO 17296-2 groups AM into seven process categories. Alternative classification: by starting material (liquid, powder, solid wire/sheet) or by energy source (laser, electron beam, thermal, UV).

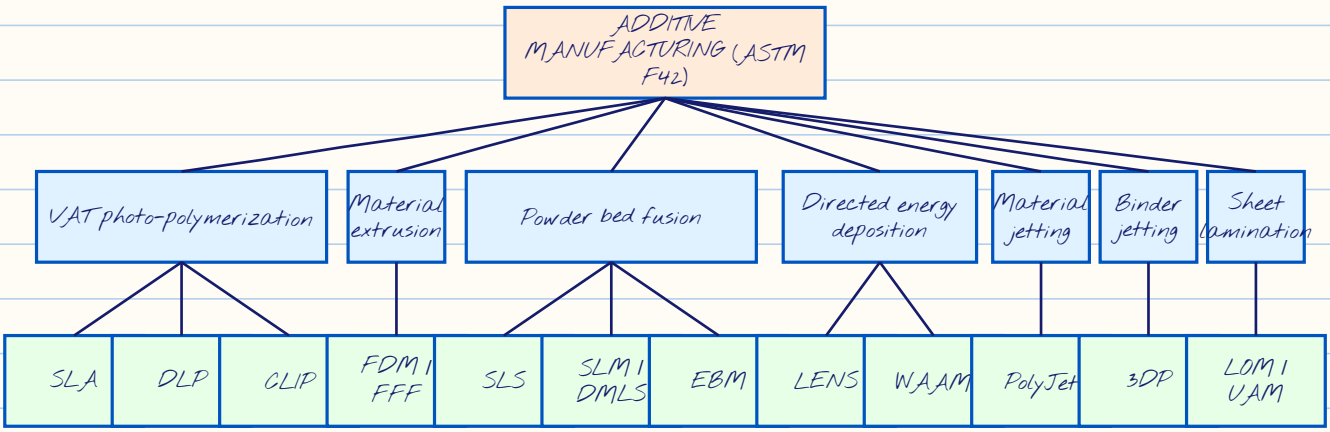


Fig 3.1 Classification of AM processes (7 categories)

Category	Energy / bonding	Material	Examples (companies)
Vat photo-polymerization	UV light cures liquid resin	Photopolymers, ceramics-filled resin	SLA (3D Systems), DLP, CLIP (Carbon)
Material extrusion	Heated filament extruded; thermal bonding	ABS, PLA, PC, PEEK, nylon	FDM (Stratasys), FFF desktop printers
Powder bed fusion	Laser / e-beam melts or sinters powder	Polymers, metals, ceramics	SLS, SLM, DMLS (EOS), EBM (Arcam)
Directed energy deposition	Focused thermal energy melts wire/powder as deposited	Metals (Ti, Ni, steels)	LENS (Optomec), WAAM, EBAM (SciaKy)
Material jetting	Droplets jetted & UV-cured	Photopolymers, wax	PolyJet, MultiJet (3D Systems)
Binder jetting	Liquid binder glues powder	Sand, metal, ceramic, gypsum	ExOne, Voxeljet, 3DP (Z Corp)
Sheet lamination	Sheets bonded (adhesive/ ultrasonic) & cut	Paper, plastic, metal foils	LOM (Helisys), UAM (Fabrisonic)

4. Advantages, Limitations & Materials

Advantages of AM

- Design freedom: complex geometry, internal channels, conformal cooling, lattices, consolidation of assemblies into one part (GE fuel nozzle: 20 parts -> 1).
- No tooling / dies - short lead time, quick design iteration, rapid prototyping,

economical for low volume.

- Mass customisation: patient-specific implants, hearing aids, dental aligners; each part can be different at no extra cost.
- Less material waste, lightweight topology-optimised parts, on-demand & distributed production (digital inventory).
- Multi-material and functionally graded parts possible; repair of expensive parts.

Limitations

- Slow build rate; part cost high for large volumes; size limited by build envelope.
- Surface finish (staircase), dimensional accuracy (+- 0.05 - 0.2 mm), anisotropy, porosity, residual stresses; limited range of qualified materials; need post-processing.
- Machine & material cost high (metal AM > 1 crore Rs.), skilled operators, safety (powders, lasers); certification & standards still developing.

Materials for AM

Class	Materials	Processes
Photopolymers (liquid resins)	Acrylates, epoxies; filled with ceramic / glass	SLA, DLP, CLIP, PolyJet
Thermoplastics	ABS, PLA, PC, PETG, nylon (PA12), PEEK, ULTEM, TPU	FDM, SLS
Metals	Ti-6Al-4V, stainless steel 316L / 17-4PH, Inconel 625/718, AlSi10Mg, Co-Cr, maraging steel, Cu, tool steels	SLM / DMLS, EBM, DED, binder jetting
Ceramics	Alumina, zirconia, silica, SiC, glass, bio-ceramics (HA)	SLA (filled), binder jetting, SLS
Composites	Carbon / glass fibre reinforced nylon, metal-polymer filaments	FDM, SLS
Others	Wax (investment patterns), sand (moulds), paper, food (chocolate), concrete, bio-ink (cells)	Binder jet, extrusion, LOM

- Powder requirements: spherical (15-45 microns) for flowability, good packing, low oxygen, narrow size distribution (gas atomised); filament: precise 1.75 / 2.85

mm dia.; resin: viscosity 50-1500 mPa s, photo-initiator.

5. VAT Photo-polymerization AM Processes

A vat of liquid photosensitive resin (photopolymer) is selectively cured by UV light / laser; monomers cross-link to a solid polymer. First commercial AM: Stereolithography (SLA).

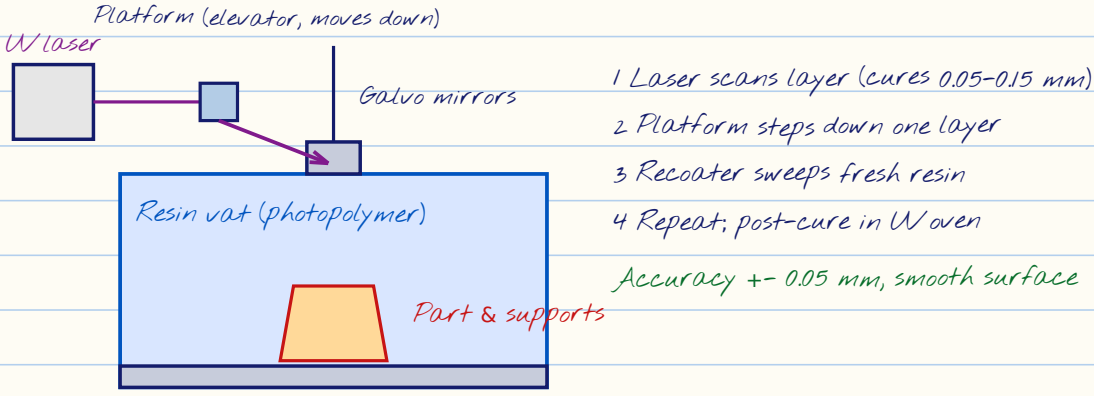


Fig 5.1 Stereolithography (SLA)

- SLA: UV laser (355 nm) draws each layer point-by-point (vector scan); platform lowers; supports are needed; part cured further in a UV oven. Layer 25-150 microns; build 250 - 2000 mm.
- DLP (Digital Light Processing): a projector cures the entire layer at once (pixel/voxel mask) - faster than SLA; resolution depends on pixels (e.g. 50 microns). LCD / MSLA: LCD mask + LED.
- CLIP / CLIP (continuous): oxygen-permeable window creates a "dead zone" so the part is pulled up continuously - 25-100x faster, smooth, isotropic parts. Bottom-up vs top-down: light from below (peel forces) / above (large vats).
- Process parameters: laser power, scan speed, layer thickness, hatch spacing, curing depth (Jacobs: $Cd = Dp \ln(E/Ec)$), resin viscosity, temperature.
- Advantages: best accuracy & surface finish among AM, fine details, large parts, transparent parts. Disadvantages: resin costly & toxic, brittle / UV-degrading parts, supports & post-curing needed, limited materials, messy. Uses: master patterns for investment casting, dental models, hearing aids, jewellery, visual prototypes, microfluidics.

6. Material Extrusion Based AM (FDM / FFF)

Fused Deposition Modelling (FDM, Stratasys 1989; FFF in open-source RepRap): thermoplastic filament is heated to semi-liquid in a nozzle and extruded along the path, bonding to the layer below by thermal fusion and solidifying quickly.

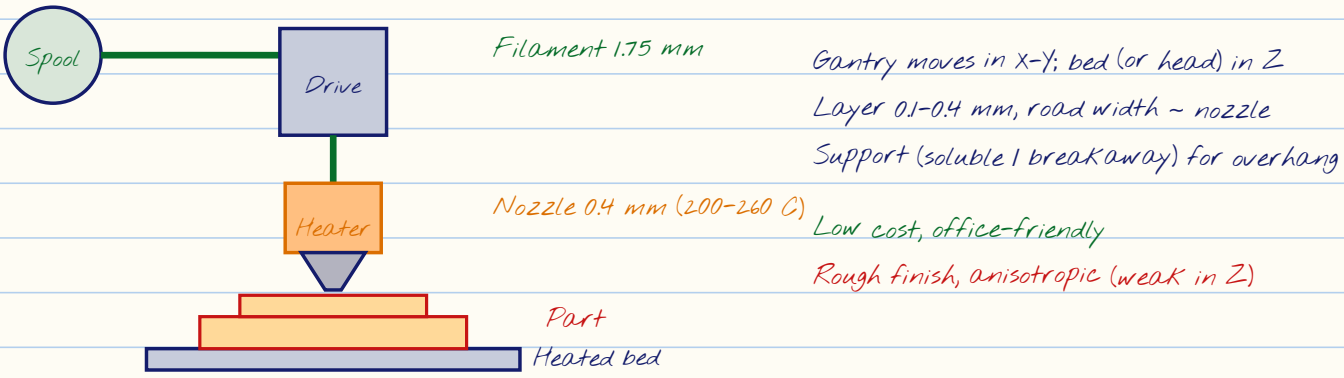
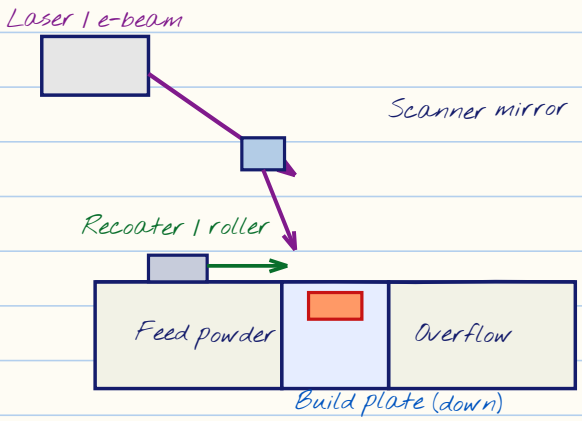


Fig 6.1 FDM / FFF process

- Parameters: nozzle & bed temperature (ABS 230/100 C; PLA 200/60 C), layer thickness, raster angle, air gap, infill % & pattern, print speed, number of contours, build orientation, cooling.
- Materials: PLA, ABS, PETG, nylon, PC, ULTEM / PEEK, TPU, carbon-fibre composites; water-soluble PVA / HIPS supports.
- Advantages: cheap machines & materials, simple, durable functional parts, multi-material, office use. Limitations: poor surface, slow for large parts, anisotropic strength (inter-layer bonds), warping/ shrinkage, visible layer lines, limited resolution.
- Defects: warping, stringing, under-extrusion, layer shifting, poor bed adhesion, voids between roads. Uses: prototypes, jigs & fixtures, tooling, drones, enclosures, education, custom prosthetics, robotics; also concrete / food / bio-printing (extrusion).

7. Powder Bed Fusion (PBF) AM Processes

PBF: a thin layer of powder is spread on the build platform by a recoater (blade / roller); a laser or electron beam selectively sinters / melts the powder along the layer contour; platform lowers; fresh powder is spread; repeat. Unfused powder supports the part (for polymers) and is reusable.



SLS: polymer powder (PA12), no support
SLM / DMLS: metal fully melted, supports
EBM: e-beam, vacuum, 700 C preheat
Inert gas (Ar / N2) for metals
Layer 20-100 microns (metal)

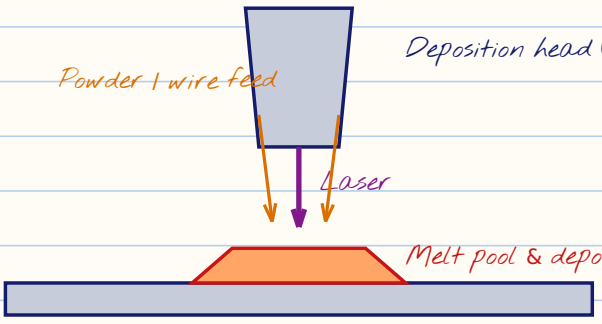
Fig 7.1 Powder bed fusion

Process	Energy / bonding	Features
SLS (Selective Laser Sintering)	CO2 laser (30-50 W) sinters polymer particles below melting (partial fusion)	PA12, TPU; no supports (powder supports); parts 95 % dense; porous
SLM / DMLS / LPBF	Fibre laser (200-1000 W) fully melts metal powder; inert Ar; supports needed	99.9 % dense; Ti, Inconel, SS, Al; heat treatment / HIP; residual stress
EBM (Electron Beam Melting)	E-beam (3-6 KW) in vacuum; bed pre-heated (~700 C+)	Less residual stress, no supports needed as much; Ti-6Al-4V, Co-Cr; rougher
Binder jet sintering (for contrast)	Binder + furnace sinter	No thermal stress, large volume, shrinkage

- Energy density: $E = P / (v \times h \times t)$ J/mm3 (P = laser power, v = scan speed, h = hatch spacing, t = layer thickness) - low E: lack of fusion / porosity; high E: Keyholing, balling, evaporation.
- Advantages: high density, fine features, mechanical properties like wrought, wide material range, no tooling, lattice & hollow parts, powder recyclable (SLS 50 % refresh).
- Limitations: costly equipment, slow, rough surface (Ra 6-15 microns), residual stresses & distortion, supports & thermal management (metal), powder safety (explosive / reactive), need HIP & machining.
- Uses: aerospace brackets / fuel nozzles, turbine blades, orthopaedic implants (porous Ti), dental crowns, heat exchangers, tooling inserts with conformal cooling, motorsport.

8. Directed Energy Deposition (DED) AM Processes

DED: a focused thermal source (laser, electron beam, plasma or arc) creates a melt pool on the substrate while metal powder or wire is fed coaxially/ side-fed into it; material is deposited and fused as the head (or workpiece) moves. Not a powder bed - deposition happens where the energy is.



LENS, DMD, WAAM, EBAM, laser cladding
Deposition rate 0.5 - 10 Kg/h (WAAM)
Large parts (metres), repairs, cladding
Poor accuracy -> machining needed
Multi-material / graded structures

Fig 8.1 Directed energy deposition

- Types: laser-based (LENS - Laser Engineered Net Shaping, DMD), WAAM (wire + arc, cheap, high rate), EBAM (electron beam wire, vacuum), plasma deposition, cold-spray (solid-state).
- Materials: Ti, Ni superalloys, stainless steels, Al, Cu, tool steels, cermets - powder or wire.
- Advantages: repair & remanufacture of turbine blades, dies; add features onto existing parts; large builds; functionally graded material; high deposition rate; use of 4-5 axis robots.
- Limitations: lower accuracy & surface finish (needs machining), thermal distortion, limited fine features, no support structures for overhangs (need multi-axis), costly.

Other AM processes (for completeness)

- Material jetting (PolyJet / MJM): inkjet heads deposit droplets of photopolymer / wax that are UV-cured at once; multi-material & full colour, very smooth (16 micron layers); gel-like support dissolved by water jet; parts less durable. Uses: realistic prototypes, medical models, overmoulds.
- Binder jetting (3DP): print head sprays liquid binder on a powder bed; green part is depowdered, then infiltrated / sintered. No heat during print - no residual stress; fast, large; sand cores & moulds, metal parts, full-colour gypsum models. Lower strength / density.
- Sheet lamination (LOM, UAM): sheets are bonded (adhesive or ultrasonic

welding) and the contour is cut by laser / blade; cheap, big parts, little distortion; waste & weak Z bond; paper, plastic, metal foil parts, embedded sensors.

Design for Additive Manufacturing (DfAM)

- Exploit freedom: topology optimisation, lattice / cellular structures, conformal cooling channels, part consolidation, organic shapes (generative design).
- Respect limits: minimum wall thickness (0.4–1 mm), minimum hole size, overhang angle < 45 degrees needs support, avoid large flat horizontal areas (warping), provide escape holes for powder / resin, add fillets, orient critical surfaces upwards, leave machining stock for critical fits.
- Support structures: block, cone, lattice, tree - provide anchor against warping and heat sink (metal), cost time & material; minimise by orientation & self-supporting angles.

Common defects in AM parts

Defect	Cause	Remedy
Warping / curling	Residual thermal stress, shrinkage, poor bed adhesion	Heated bed, brim / raft, supports, preheat, stress relief
Porosity (gas / lack of fusion)	Low energy density, wet powder, trapped gas, bad hatch	Optimise power / speed / hatch, dry powder, HIP
Balling / keyholing (metal)	Too high speed / power; unstable melt pool	Adjust energy density, scan strategy
Cracking / delamination	Thermal gradients, poor layer bonding	Preheat, reduce layer cooling, better parameters
Staircase / poor finish	Layer thickness, orientation	Thinner layers, orientation, post finishing
Dimensional inaccuracy	Shrinkage, stair effects, calibration	Scale compensation, calibration, post-machining

Energy / process parameters - quick formulae

- PBF volumetric energy density $E = P / (v\ h\ t)$ [J/mm^3]; build time = (volume / build rate) + recoating time per layer × layers.
- Number of layers n = part height / layer thickness; FDM extrusion volume rate = road width × layer height × speed.

- SLA cure depth $C_d = D_p \ln(E / E_c)$ (D_p = penetration depth, E_c = critical exposure).

Solved example - FDM build time estimate

Part 60 mm tall printed at 0.2 mm layers -> $n = 300$ layers. If each layer takes 4 min (perimeter + infill), build time = 300 x 4 = 1200 min = 20 hours. Doubling layer height to 0.4 mm halves the layers (~ 10 h) but staircase worsens.

9. Post-Processing of AM Parts

Raw AM parts rarely meet final needs; post-processing improves surface finish, dimensional accuracy, strength and removes supports.

Operation	Purpose / methods	Applicable to
Cleaning & depowdering	Remove loose powder (brushing, air, ultrasonic), resin washing in IPA	PBF, SLA, binder jet
Support removal	Break-away, cutting, wire EDM, soluble dissolving (water / chemical), machining	FDM, SLA, metal PBF
Post-curing	UV / thermal cure to full strength	SLA, DLP, PolyJet
Heat treatment / stress relief	Relieve residual stress (eg. 650-800 C for Ti), age hardening, solutionise	Metal AM
HIP (hot isostatic pressing)	Heat + 100-200 MPa gas closes internal pores -> 100 % density, improved fatigue	Metal AM, critical parts
Machining / grinding / EDM	Tight tolerances, threads, bearing seats, datum faces	Metal & DED parts
Surface finishing	Sanding, bead / sand blasting, tumbling, polishing, electro-polishing, chemical / vapour smoothing (acetone for ABS), barrel finishing, coatings, painting, plating	All
Infiltration	Fill pores with epoxy, wax, bronze (binder jet metal), improving strength	Binder jet, SLS
Sintering / debinding	Remove binder & sinter green part in furnace (shrinks 15-20 %)	Binder jet, metal FFF
Inspection	CMM, 3D scan, CT, dye penetrant, density test	All

10. Applications of AM

Sector	Applications
Aerospace & defence	Lightweight brackets, fuel nozzles (LEAP), turbine blades, ducts, satellite parts, UAV frames, repair of blades; Ti & Ni alloys
Automotive	Prototypes, jigs & fixtures, intake manifolds, lightweight custom parts, motorsport components, tooling with conformal cooling
Medical & dental	Patient-specific implants (hip, cranial, spine), surgical guides, prosthetics, hearing aids, dental crowns & aligners, anatomical models for surgical planning, bioprinting of tissues
Tooling & manufacturing	Injection-mould inserts, die-casting cores, sand moulds / cores (binder jet), investment patterns, end-of-arm grippers, rapid tooling
Consumer & art	Jewellery, eyewear, footwear midsoles, fashion, sculptures, toys, customised products
Construction & others	Concrete 3D printed houses, bridges; food printing; education; energy (heat exchangers); electronics (printed circuits); spare parts on demand

Trends & emerging areas

- Bioprinting: living cells + hydrogel bio-ink to print tissue scaffolds, skin, cartilage.
- 4D printing: printed structure that changes shape / property with time under stimulus (heat, water, light).
- Large-format & construction printing, hybrid AM-CNC machines, in-situ monitoring with AI, multi-laser machines, sustainable (recycled powder), on-demand spare parts, space manufacturing.
- Standards: ASTM F42, ISO / ASTM 52900 (terminology); qualification of aerospace and medical parts is key.

Quick viva / 2-mark questions

- What is STL? -> Standard Tessellation Language: surface model approximated with triangles.
- Name the 7 ASTM categories. -> VPP, ME, PBF, DED, MJ, BJ, SL.
- Why supports? -> To hold overhangs, anchor part and conduct heat. What is HIP? -> Hot isostatic pressing to remove pores.
- Difference SLS vs SLM? -> Sintering (partial fusion, polymers) vs full melting

(metals, dense).

- Which AM suits repair of a turbine blade? -> DED (laser cladding).

Process comparison - quick revision

- SLA / DLP: liquid resin + UV; best finish & accuracy; supports; brittle.
- FDM: filament extrusion; cheap, strong plastics; layer lines; anisotropy.
- SLS / SLM / EBM: powder + laser / e-beam; dense functional parts; metals; supports (metal); costly.
- DED: wire / powder + focused energy; large parts, repair; poor finish.
- Trade-off always: speed vs accuracy vs cost vs material range.

Important exam questions - Unit V

- Explain the steps involved in additive manufacturing with a neat chain diagram. (10 M)
- Classify AM processes (ASTM F42). Advantages & limitations of AM. Materials used. (10 M)
- VAT photo-polymerization (SLA / DLP) - process, parameters, merits & demerits. (10 M)
- FDM / extrusion-based AM with sketch; applications. (10 M)
- Powder bed fusion (SLS, SLM, EBM); DED processes. (10 M each)
- Post-processing of AM parts; applications of AM in different fields. (5-10 M)